

Dexter Fernandes

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AI Engineer

WORK EXPERIENCE

Esbaar • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Designed and deployed real-time CNN perception pipelines on edge accelerators using PyTorch and OpenVINO, achieving a 15% F1 improvement with deterministic 24/7 operation under harsh industrial conditions.
- Built an automated sensor-data cleansing pipeline handling 3.6 million records, reducing model rollout time from 3 days to under 24 hours.
- Implemented automated data ingestion, cleaning, training, and validation pipelines for large multimodal sensor datasets using MLflow, DVC, and Optuna, streamlining experimentation and reducing model iteration time by 40%.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.
- Owned end-to-end data pipelines, including automated data collection, augmentation, and labelling, increasing dataset diversity by 40% and reducing manual labelling effort by 60%.
- Documented deployment guardrails, failure modes, and recovery procedures, reducing cognitive load for engineers operating and maintaining fielded systems.
- Collaborated with cross-functional engineering and client teams to translate performance metrics into actionable deployment and optimisation decisions.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI • Full-time

- Led a 4-member team to design, train, and deploy CNN-based models for real-time object detection, pose estimation, and tracking, reducing false positives by 33% and achieving sub-50 ms inference latency on edge hardware.
- Deployed quantised vision models to rugged Intel VPU edge devices using OpenVINO, delivering deterministic real-time inference across RGB and IR sensor streams in harsh industrial environments.
- Re-architected the embedded inference pipeline and model optimisation workflow, increasing latency from 5 FPS to 15 FPS, enabling real-time detection of safety-critical events.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60%.
- Mentored engineers and collaborated with hardware, software, and QA teams to integrate perception modules into deployed systems and improve delivery consistency.
- Presented key findings and metrics to leadership and clients, translating complex performance results into actionable insights for deployment and further optimisation.
- Designed data handling and retention practices aligned with GDPR principles, including data minimisation and controlled access, within production computer vision systems.
- Owned the full machine learning pipeline from data ingestion and labelling through training, optimisation, deployment, monitoring, and field validation, accelerating iteration cycles and improving deployment reliability.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system integrating LiDAR and radiological data for the teleoperation of the Clearpath Husky UGV development platform.
- Implemented ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration.
- Developed and maintained ROS2 nodes and interfaces integrating perception outputs into a broader robotic system, managing sensor topics, tf2 frames, and message synchronisation for downstream localisation and navigation.
- Integrated perception pipelines into ROS-based robotic workflows, validating end-to-end data flow from sensors to SLAM and navigation components in Gazebo prior to deployment.
- Simulated and validated 2D/3D mapping and navigation workflows in Gazebo and ROS2 prior to real-world deployment.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Strong foundation in signals, systems, embedded hardware, and control, supporting later work in real-time AI and perception systems.

CERTIFICATIONS

Deep Learning Specialisation

Jul 2019

deeplearning.ai

AWARDS & SCHOLARSHIPS

Best Speaker award

Jan 2020

ABC Toastmasters club

PROJECTS

ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulated mobile robot integrating 3D LiDAR and RGB camera data in Gazebo Harmonic to perform real-time SLAM and autonomous navigation using ROS 2 Jazzy.
- Applied slam_toolbox and Nav2 for 2D/3D mapping, localisation, and waypoint-based navigation with optimised Ceres backends.
- Processed multi-sensor data streams for spatial alignment and tested mapping accuracy across diverse simulated environments.
- Developed modular ROS 2 launch files and parameterised configurations for flexible experimentation with perception and navigation pipelines.
- Containerised the full application using Docker to ensure reproducibility, version control, and consistent deployment across development environments.

SKILLS

- **Perception & SLAM:** Convolutional Neural Networks, Data Augmentation, NumPy, Object Detection & Tracking, OpenCV, Scikit-Image, Scikit-Learn, U-Net, Vision Transformer (ViT), YOLO
- **Cloud, Edge & MLOps:** Apache Kafka, AWS, Docker, DVC (Data Version Control), FastAPI, Microsoft Azure, MLFlow
- **Programming & Development:** Bash, C++17, CMake, Git, Linux, Pandas, PostgreSQL, Python

- **Deep Learning and Optimisation:** Bayesian Optimisation, Layer fusion, MultiRocket, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Structured pruning, TensorFlow, TensorRT
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership